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阅读 C

动脑难，难在违背大脑的习惯

说明文 · 脑科学与学习方法 · 全文 468 词 · 逐句精读

¹ For **decades**, scientists **hypothesized** that **cognitively demanding** tasks, such as writing or solving math problems, **required** more **energy** than easier tasks like **scrolling** short videos on the phone or looking out of the window.

几十年来，科学家们一直假设，写作或解数学题这类高认知要求的任务，比刷手机短视频或望着窗外这类轻松的任务更耗能量。

decade n. 十年

hypothesize v. 假设

cognitively adv. 在认知上

demanding adj. 要求高的；费力的

require v. 需要

energy n. 能量

scroll v. 滚动浏览；刷（屏）

² However, **recent** research **proves** the **view** wrong.

然而，最近的研究证明这种观点是错误的。

recent adj. 最近的

prove v. 证明

view n. 观点

³ Our brains **perform** **massively** **complex** and **metabolically** **costly** **computations** **all day long**.

我们的大脑整天都在进行极其复杂、代谢成本很高的运算。

perform v. 执行；进行

massively adv. 极其；非常

complex adj. 复杂的

metabolically adv. 在新陈代谢方面

costly adj. 代价高的

computation n. 计算；运算

all day long 一整天

⁴ “Whether you’re **engaged in** a cognitively demanding task at work or you’re **staring off into space**, your brain is using **roughly** the same **amount** of energy,” says **neuroscientist** Todd Braver of Washington University in St. Louis.

“无论你是否在工作中忙于一项费脑的任务，还是在发呆，你的大脑消耗的能量都大致相同，” 圣路易斯华盛顿大学的神经科学家托德·布雷弗说。

be engaged in 忙于；从事

stare off into space 发呆

roughly adv. 大约

amount n. 数量

neuroscientist n. 神经科学家

⁵ In fact, a task feels more challenging when it requires the brain to go against its usual habits.

事实上，一项任务如果要求大脑违背其通常的习惯，就会让人觉得更有挑战性。

in fact 事实上

challenging adj. 有挑战性的

go against 违背

usual adj. 通常的

habit n. 习惯

⁶ Cognitively demanding work often requires the **coordination** of many different brain **regions** in **novel** **patterns** and the **silencing** of **circuits** that **frequently** **fire** together.

高认知要求的工作，常常需要大脑许多不同区域以新的模式协同运作，同时需要让那些经常一起被激活的神经回路安静下来。

coordination n. 协调

region n. 区域

novel adj. 新颖的

pattern n. 模式

silence v. 使安静；使沉寂（此处用动名词 the silencing of）

circuit n. 回路；电路

frequently adv. 频繁地

fire v. （神经元）放电；激活

⁷ Studies have found that pushing against our brain's habits triggers an emotional response, making us feel uncomfortable.

研究发现，违背大脑的习惯会引发一种情绪反应，让我们感到不舒服。

push against 对抗；违背

trigger v. 引发

emotional adj. 情绪的

response n. 反应

uncomfortable adj. 不舒服的

⁸ “It’s **aversive**, and it just feels bad,” says Michael Inzlicht, a **psychologist** from the University of Toronto.

“这让人反感，感觉就是很糟糕，” 多伦多大学的心理学家迈克尔·因兹利希特说。

aversive adj. 令人厌恶的；令人反感的

psychologist n. 心理学家

⁹ But he **points out** that emotional responses are **subjective**: they are **interpretations** of **sensations** in our bodies and brains.

但他指出，情绪反应是主观的：它们是对我们身体和大脑中各种感觉的解读。

point out 指出

subjective adj. 主观的

interpretation n. 解读；理解

sensation n. 感觉

¹⁰ And we can **influence** how much we **pay attention to** these sensations and even how we **interpret** them.

而我们可以影响自己对这些感觉的关注程度，甚至可以影响我们如何解读它们。

influence v. 影响

pay attention to 注意

interpret v. 解释；理解

¹¹ We often **overestimate** the **struggle** and pain required to **complete** **cognitive** **tasks**.

我们常常高估完成认知任务所需的艰辛和痛苦。

overestimate v. 高估

struggle n. 艰辛；费力的事

complete v. 完成

cognitive adj. 认知的

task n. 任务

¹² This **initial** **overestimation** can **lead us to** **procrastinate**,
or even **avoid** the task **altogether**.

这种最初的高估可能导致我们拖延，甚至完全回避这项任务。

initial adj. 最初的

overestimation n. 高估

lead sb to do sth 促使某人做某事；导致某人做某事

procrastinate v. 拖延

avoid v. 回避

altogether adv. 完全

¹³ Studies **suggest** that if we can **get past** that initial **hurdle** of starting a task, then thinking becomes easier.

研究表明，如果我们能越过“开始一项任务”这道最初的障碍，思考就会变得更加容易。

suggest v. 表明

get past 越过

hurdle n. 障碍；跨栏

¹⁴ One way to do that is to pay attention to the **benefits** **instead of** the **effort** required.

做到这一点的一个办法是把注意力放在好处上，而不是放在所需付出的努力上。

benefit n. 好处；益处

instead of 而不是

effort n. 努力

- 15 In a study published in the journal Science in 2020, Andrew Westbrook, a neuroscientist at Rutgers University, and his colleagues gave participants the choice of solving an easy memory puzzle for a small amount of money or a much harder puzzle for more money.

在2020年发表于《科学》杂志的一项研究中，罗格斯大学的神经科学家安德鲁·韦斯特布鲁克和同事让参与者选择：解一道简单的记忆题赚一小笔钱，或者解一道难得多的题赚更多钱。

publish v. 发表

journal n. 期刊

colleague n. 同事

participant n. 参与者

choice n. 选择

puzzle n. 谜题；难题

¹⁶ The **options** were **displayed** on a screen, and participants' eye **movements** were **tracked** as they decided which puzzle to **attempt**.

这些选项显示在屏幕上，在参与者决定尝试哪道题时，研究人员追踪了他们的眼球运动。

option n. 选项

display v. 显示

movement n. 运动

track v. 追踪

attempt v. 尝试

¹⁷ When people spent more time looking at the **reward** for the challenging puzzle, they were more **likely to** choose it.

当人们花更多时间去看那道难题的奖励时，他们就更有可能会选择它。

reward n. 奖励

be likely to 可能

¹⁸ If our **mind's eye**, or our **attentional focus**, is on the benefits of an option, **over time** we're more likely to choose to do hard things.

如果我们的“心灵之眼”，也就是注意力的焦点，放在某个选项的好处上，那么随着时间推移，我们就更有可能选择去做困难的事。

mind's eye 心灵之眼（此处指注意力的焦点）

attentional adj. 注意力的

focus n. 焦点

over time 随着时间推移

¹⁹ Performing complex **mental** tasks will also be less **painful** if the **exertion** of mental **effort** becomes **habitual**.

如果付出脑力成了一种习惯，执行复杂的脑力任务也就不那么痛苦了。

mental adj. 精神的；脑力的

painful adj. 痛苦的

exertion n. 付出；运用（力气、脑力等）

effort n. 努力；气力

habitual adj. 习惯性的；成为习惯的

²⁰ When we don't have to **make a decision** to **focus** every day, we will save ourselves considering the costs or **consequences**.

当我们不必每天都下决心去专注时，就省得去考虑代价或后果了。

make a decision 做决定

focus v. 集中注意力

consequence n. 后果

²¹ To make cognitively demanding tasks a habit, try to do them every day at the same time and in the same place.

要让高认知要求的任务成为习惯，试着每天在同一时间、同一地点去做。

at the same time 在同一时间

²² Add a **ritual** before the work starts, such as **turning off** the phone and **launching** an app to **block** **distracting** websites.

在工作开始前加一个仪式，比如关掉手机，打开一个应用来屏蔽让人分心的网站。

ritual n. 仪式；惯例

turn off 关掉

launch v. 启动；开启

block v. 阻止；屏蔽

distracting adj. 使人分心的

²³ After a few weeks, concentrating deeply won't just feel easier; it will actually be easier.

几周之后，深度专注不仅会感觉更容易，而且真的会变得更加容易。

concentrate v. 集中注意力

deeply adv. 深入地

actually adv. 实际上